

Demo - Decision Tree using Python

In this demo, you will learn how to apply decision tree algorithm in Python using scikit-learn library.

Problem Statement

The dataset contains information about 3 brands namely US, Japan, Europe. Other parameters are mpg, cylinders, cubicinches, hp, weightlbs, time-to-60, year. The task is to find the brand of a car using these parameters.

Importing Required Libraries

```
In [ ]: import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.tree import plot_tree
from sklearn.metrics import accuracy_score
```

Dataset

```
In [ ]: #Reading the dataset using pandas
car_data=pd.read_csv('cars.csv')
car_data.head()
```

Data Analysis

```
In [ ]: #Dataset shape to get row and column count
car_data.shape
```

- Checking Null Values

```
In [ ]: #Checking for any null values
car_data.isnull().any()
```



None of the columns contain null values so there is no need to remove/replace data



• Encoding Target Variable

```
In [ ]: #Using LabelEncoder to encode the target variable
lb=LabelEncoder()
lb.fit_transform(car_data['brand'])
```

• Divide Feature And Target Variables

```
In [ ]: #Divide the dataset into independent and dependent variables
X=car_data.drop('brand',axis=1)
y=car_data['brand']
```

• Training And Testing Set

```
In [ ]: #Split the data into training and testing set
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2) #20 percent of data
```

Decision Tree Training

The following parameters should be considered during training a decision tree:

- **criterion:** To measure quality of a split. Supported criteria are "gini" for the Gini impurity and "entropy" for the information gain.
- **splitter:** Either 'best' to choose best split or 'random' to choose random best split. By default takes 'best' value.
- **max_features:** Number of features to consider.
- **max_depth:** The maximum depth of the tree. If None, then nodes are expanded until all leaves are pure or until all leaves contain less than min_samples_split samples.

The classifier has several parameter which are used depending on the requirement. For more information about these parameters you can visit: [Decision Tree](#)

```
In [ ]: dt_car=DecisionTreeClassifier(criterion='entropy',splitter='best',max_features=3,max_c
dt_car.fit(X_train,y_train)
```

```
In [ ]: #plot tree can also be used for visualising the tree
plt.figure(figsize=(20,20))
plot_tree(dt_car,class_names=list(lb.classes_),feature_names=list(X_train.columns))
plt.show()
```

```
In [ ]: #Calculating accuracy
pred = dt_car.predict(X_test)
print('Accuracy is:',accuracy_score(y_test,pred))
```

```
In [ ]:
```